

1 Theory

A flip-flop is a fundamental building block in digital electronics, used for storing a single bit of binary data. It has two stable states, typically represented as 0 and 1, and can be controlled using input signals and a clock pulse. Flip-flops are the basis for memory elements in sequential circuits, enabling the storage, transfer, and synchronization of data. There are several types of flip-flops, such as T, D, SR and JK; each with specific input-output behaviors suited for different applications, including data storage, counters, and state machines.

T Flip-Flop

The T flip-flop (Toggle Flip-Flop) toggles its output state on every clock cycle if the input T is HIGH(1). If T is LOW(0), the output remains unchanged, retaining its previous state. It is primarily used in counters and frequency division circuits due to its toggle behavior.

Characteristic Table

T	Q	$\bar{Q}$
0	0	0
1	0	1
0	1	1
1	1	0

D Flip-Flop

The D flip-flop transfers the input D to the output Q on the triggering edge of the clock signal. This ensures the output matches the input and eliminates undefined conditions. Its simplicity makes it ideal for data storage and shift registers.

Characteristic Table

D	Q	$\bar{Q}$
0	0	1
0	0	1
1	0	1
1	1	0

SR Flip-Flop

The SR flip-flop (Set-Reset Flip-Flop) uses two inputs, S(Set) and R(Reset), to control the output Q. When S=1, the output is set to 1, and when R=1, the output is reset to 0. The condition S=R=0 holds the previous state, while S=R=1 is undefined.

Characteristic Table

S	R	Q	$\bar{Q}$
0	0	0	0
0	1	0	0
1	0	0	1
1	1	0	-
0	0	1	1
0	1	1	0
1	0	1	1
1	1	1	-

JK Flip-Flop

The JK flip-flop is an enhanced version of the SR flip-flop, resolving its undefined condition. It has two inputs, J and K. When both inputs are HIGH, the output toggles between 0 and 1. This makes it versatile for use in counters, registers, and sequential circuits.

Characteristic Table

J	K	Q	$\bar{Q}$
0	0	0	0
0	1	0	0
1	0	0	1
1	1	0	1
0	0	1	1
0	1	1	0
1	0	1	1
1	1	1	0

2 Implementation and Observation

2.1 T Flip-Flop

VHDL code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity flipflop is
    Port ( T : in STD_LOGIC;
          CLK : in STD_LOGIC;
          QN : in STD_LOGIC;
          QOUT : out STD_LOGIC);
end flipflop;
```

architecture Behavioral of flipflop is

```
begin
process(T,CLK,QN)is
variable temp:std_logic:='0';
begin
if (QN='1')then
temp:='0';
elsif(rising_edge(clk))then
if (T='1')then
temp:= not temp;
end if;
end if;
QOUT<= temp;
end process;
end Behavioral;
```

Test Bench

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity flipflop_tb is
end flipflop_tb;

architecture Behavioral of flipflop_tb is
    component flipflop port( T : in STD_LOGIC;
        QN : in STD_LOGIC;
        CLK : in STD_LOGIC;
        QOUT : out STD_LOGIC);
    end component;
    signal T :STD_LOGIC;
    signal QN :STD_LOGIC;
    signal CLK :STD_LOGIC:= '0';
    signal QOUT :STD_LOGIC;

begin
    UUT: flipflop port map(
    T=>T,
    QN=>QN,
    CLK=>CLK,
    QOUT=>QOUT);
    process
    begin
        clk<=not clk;
        wait for 10ns;
```

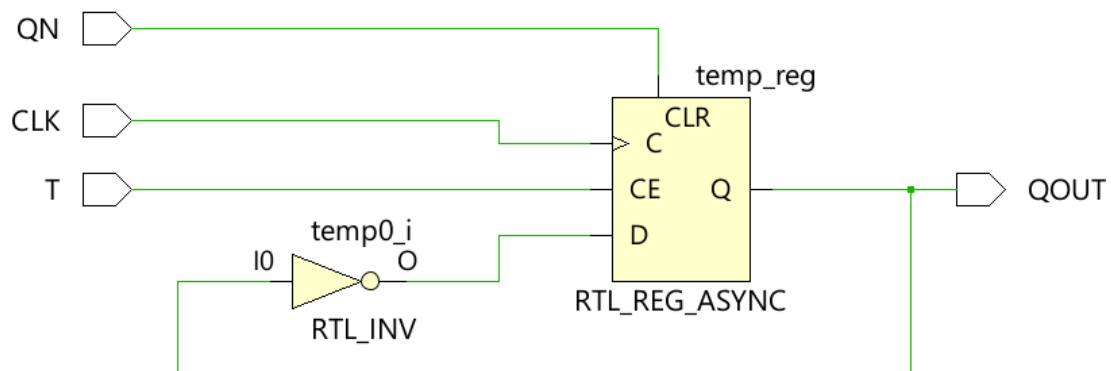


Figure 1: RTL Schematic Diagram of t flipflop

```

end process;
process
begin
T<='0';
wait for 10ns;
T<='1';
wait for 10ns;
T<='1';
wait;
end process;
end Behavioral;

```

RTL Schematic Diagram

RTL design is used to describe the behavior of the t flipflop as shown in figure 1

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the t flipflop circuit as shown in figure 2

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table of t flipflop as shown in figure 3.

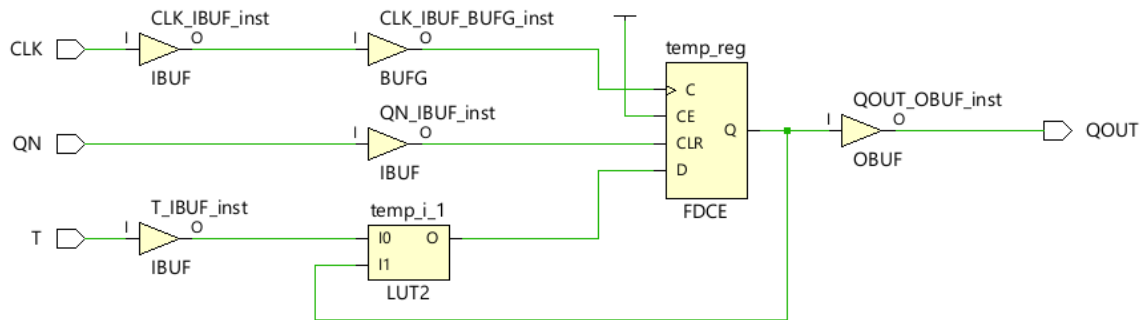


Figure 2: Synthesis Schematic Diagram of t flipflop

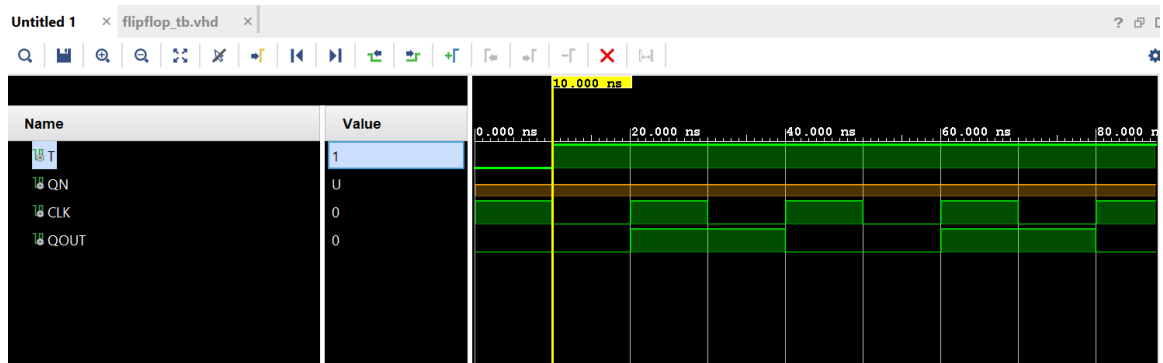


Figure 3: Timing Diagram of t flipflop

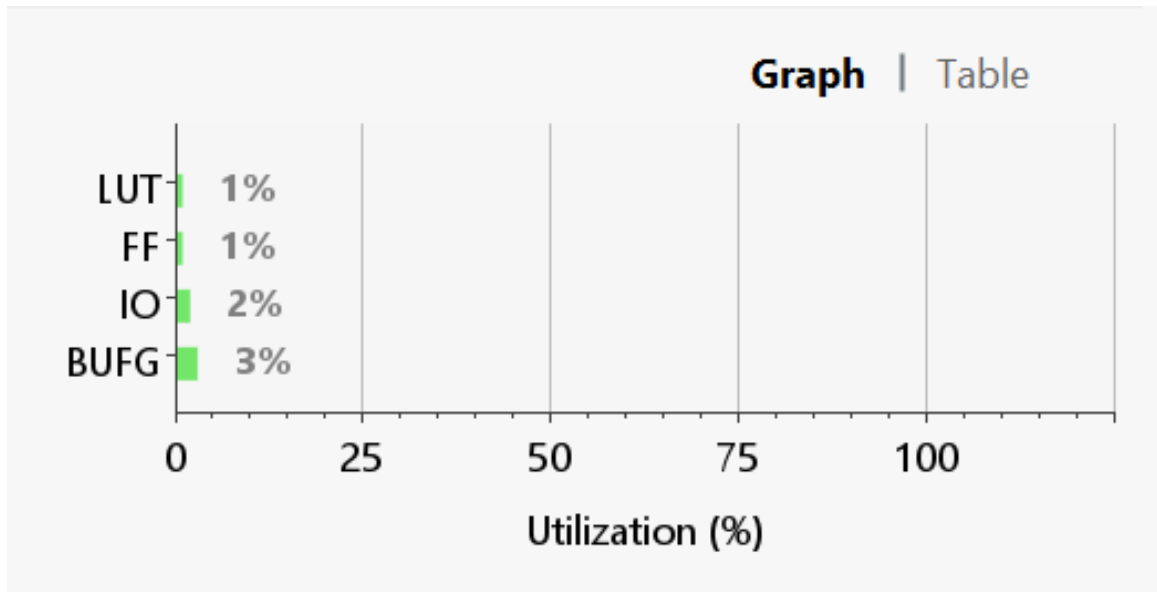


Figure 4: Device Utilization of t flipflop

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 4.

Power Estimation

Figure 5 shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a t flipflop circuit.

2.2 D Flip-Flop

VHDL code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity DFLIPFLOP is
    Port ( DIN : in STD_LOGIC;
          CLK : in STD_LOGIC;
          RESET : in STD_LOGIC;
          DOUT : out STD_LOGIC);
end DFLIPFLOP;
```

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: 1.062 W
Design Power Budget: Not Specified
Process: typical
Power Budget Margin: N/A
Junction Temperature: 30.1°C
Thermal Margin: 54.9°C (11.4 W)
Ambient Temperature: 25.0 °C
Effective θ_{JA} : 4.8°C/W
Power supplied to off-chip devices: 0 W
Confidence level: Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power

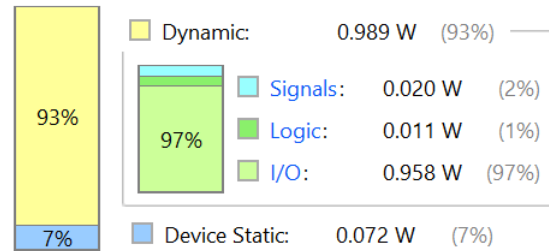


Figure 5: Power Estimation of t flipflop

architecture Behavioral of DFLIPFLOP is

```
begin
DFLIPFLOP:process(DIN,CLK,RESET)is
begin
if (RESET='1')then
DOUT<='0';
elsif(rising_edge(clk))then
DOUT<= DIN;
end if;
end process DFLIPFLOP;
end Behavioral;
```

Test Bench

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity DFLIPFLOP_tb is
end DFLIPFLOP_tb;

architecture Behavioral of DFLIPFLOP_tb is
component DFLIPFLOP port( DIN : in STD_LOGIC;
```

```

        RESET : in STD_LOGIC;
        CLK : in STD_LOGIC;
        DOUT : out STD_LOGIC);
end component;
    signal DIN :STD_LOGIC;
    signal RESET :STD_LOGIC;
    signal CLK :STD_LOGIC:= '0';
    signal DOUT :STD_LOGIC;

begin
UUT: DFLIPFLOP port map(
DIN=>DIN,
RESET=>RESET,
CLK=>CLK,
DOUT=>DOUT);
process
begin
clk<=not clk;
wait for 10ns;
end process;
process
begin
DIN<='0';
wait for 10ns;
DIN<='1';
wait for 10ns;
DIN<='1';
wait;
end process;
end Behavioral;

```

RTL Schematic Diagram

RTL design is used to describe the behavior of the d flipflop as shown in figure 6

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the d flipflop circuit as shown in figure 7

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table of d flipflop as shown in figure 8.

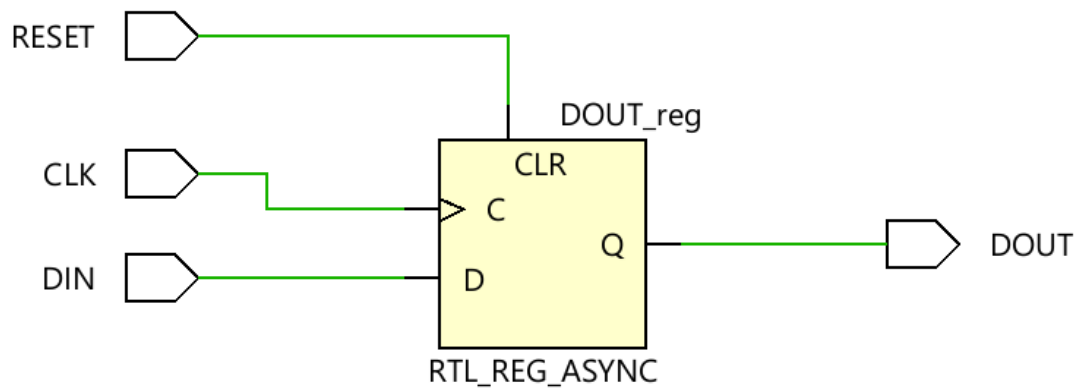


Figure 6: RTL Schematic Diagram of d flipflop

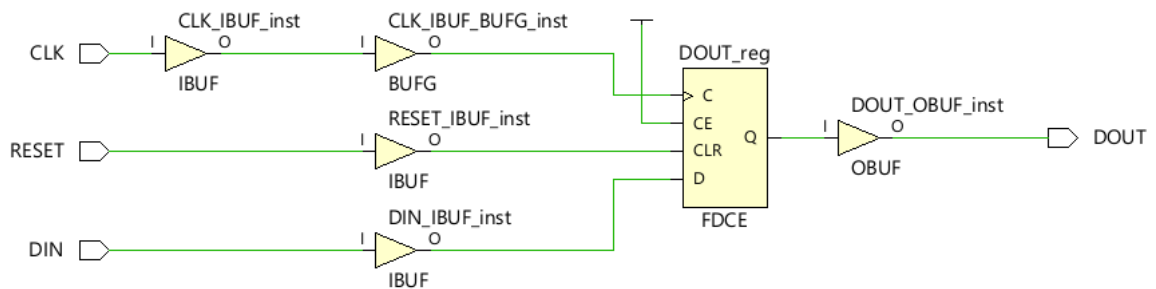


Figure 7: Synthesis Schematic Diagram of d flipflop

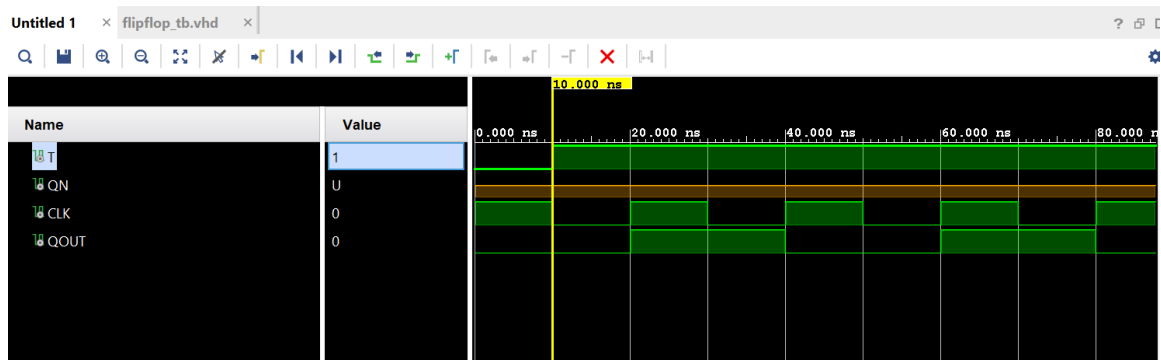


Figure 8: Timing Diagram of d flipflop

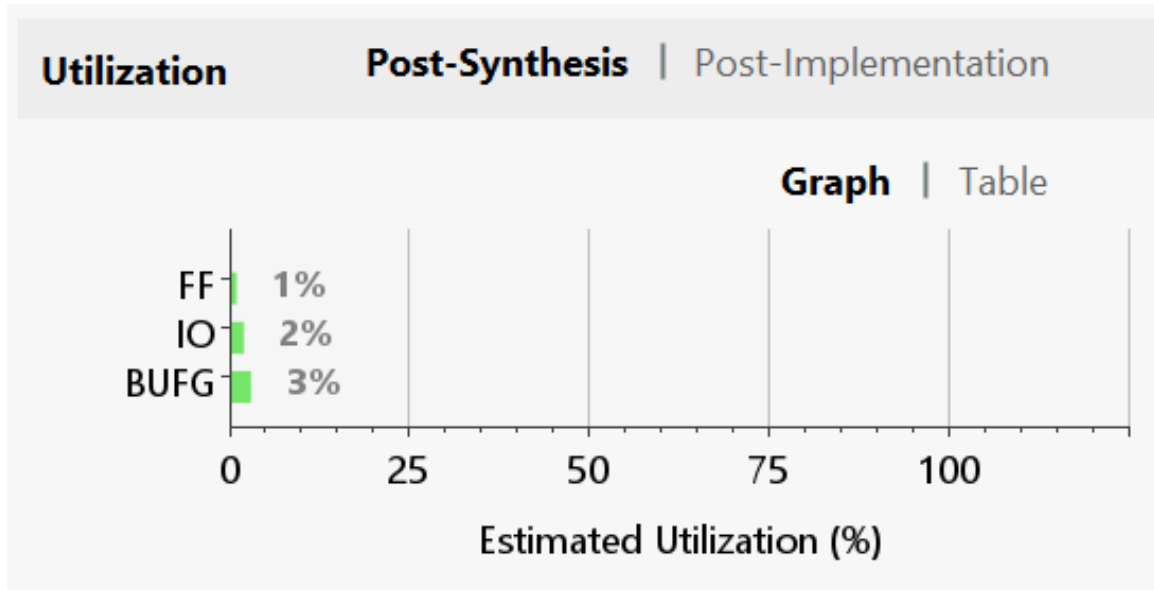


Figure 9: Device Utilization of d flipflop

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 9.

Power Estimation

Figure 10 shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a d flipflop circuit.

2.3 SR Flip-Flop

VHDL code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity SRFLIPFLOP is
    Port ( S : in STD_LOGIC;
          R : in STD_LOGIC;
          CLK : in STD_LOGIC;
          RESET : in STD_LOGIC;
          Q : out STD_LOGIC);
end SRFLIPFLOP;
```

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: 0.333 W
Design Power Budget: Not Specified
Process: typical
Power Budget Margin: N/A
Junction Temperature: 26.6°C
Thermal Margin: 58.4°C (12.2 W)
Ambient Temperature: 25.0 °C
Effective θ_{JA} : 4.8°C/W
Power supplied to off-chip devices: 0 W
Confidence level: Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power

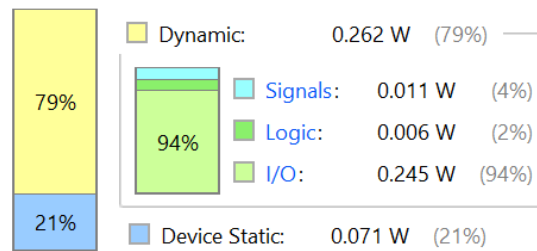


Figure 10: Power Estimation of d flipflop

architecture Behavioral of SRFLIPFLOP is

```
begin
```

```
  SR_FLIPFLOP: process(S, R, CLK, RESET)
```

```
    begin
```

```
      if (RESET = '1') then
```

```
        Q <= '0';
```

```
      elsif (rising_edge(CLK)) then
```

```
        if (S = '1' and R = '0') then
```

```
          Q <= '1';
```

```
        elsif (S = '0' and R = '1') then
```

```
          Q <= '0';
```

```
        elsif (S = '1' and R = '1') then
```

```
          Q <= 'X'; -- Invalid state
```

```
        end if;
```

```
      end if;
```

```
    end process SR_FLIPFLOP;
```

```
end Behavioral;
```

Test Bench

```
library IEEE;
```

```

use IEEE.STD_LOGIC_1164.ALL;
entity SRFLIPFLOP_tb is
end SRFLIPFLOP_tb;
architecture Behavioral of SRFLIPFLOP_tb is
    component SRFLIPFLOP
        port ( S : in STD_LOGIC;
              R : in STD_LOGIC;
              CLK : in STD_LOGIC;
              RESET : in STD_LOGIC;
              Q : out STD_LOGIC);
    end component;

    signal S, R : STD_LOGIC := '0';
    signal RESET : STD_LOGIC := '0';
    signal CLK : STD_LOGIC := '0';
    signal Q : STD_LOGIC;

begin
    UUT: SRFLIPFLOP port map (
        S => S,
        R => R,
        CLK => CLK,
        RESET => RESET,
        Q => Q
    );
    CLK_process: process
    begin
        CLK <= not CLK;
        wait for 10 ns;
    end process;
    Stimulus: process
    begin
        RESET <= '1';
        wait for 20 ns;
        RESET <= '0';
        S <= '1'; R <= '0'; wait for 20 ns; -- Set
        S <= '0'; R <= '1'; wait for 20 ns; -- Reset
        S <= '1'; R <= '1'; wait for 20 ns; -- Invalid state
        S <= '0'; R <= '0'; wait for 20 ns; -- No change
        wait;
    end process;
end Behavioral;

```

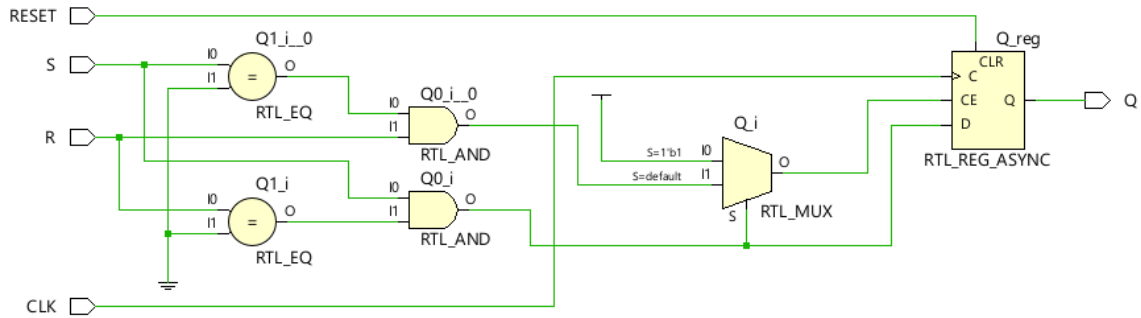


Figure 11: RTL Schematic Diagram of sr flipflop

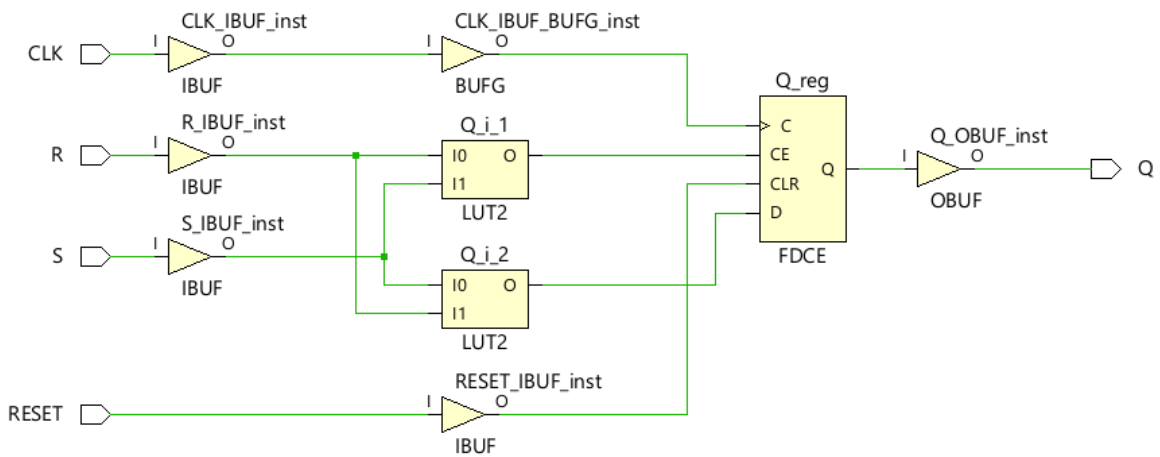


Figure 12: Synthesis Schematic Diagram of sr flipflop

RTL Schematic Diagram

RTL design is used to describe the behavior of the sr flipflop as shown in figure 11

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the sr flipflop circuit as shown in figure 12

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table of sr flipflop as shown in figure 13.

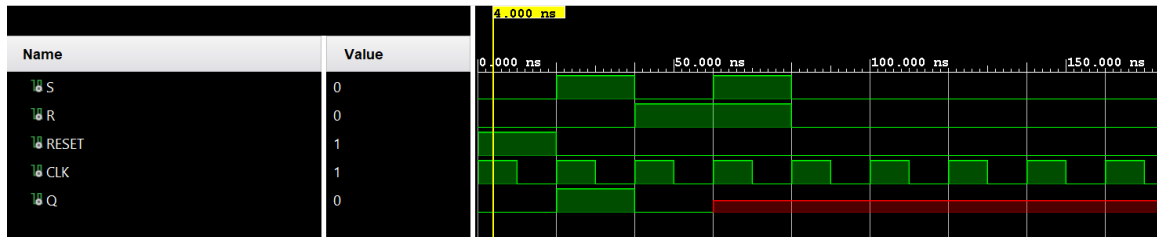


Figure 13: Timing Diagram of sr flipflop

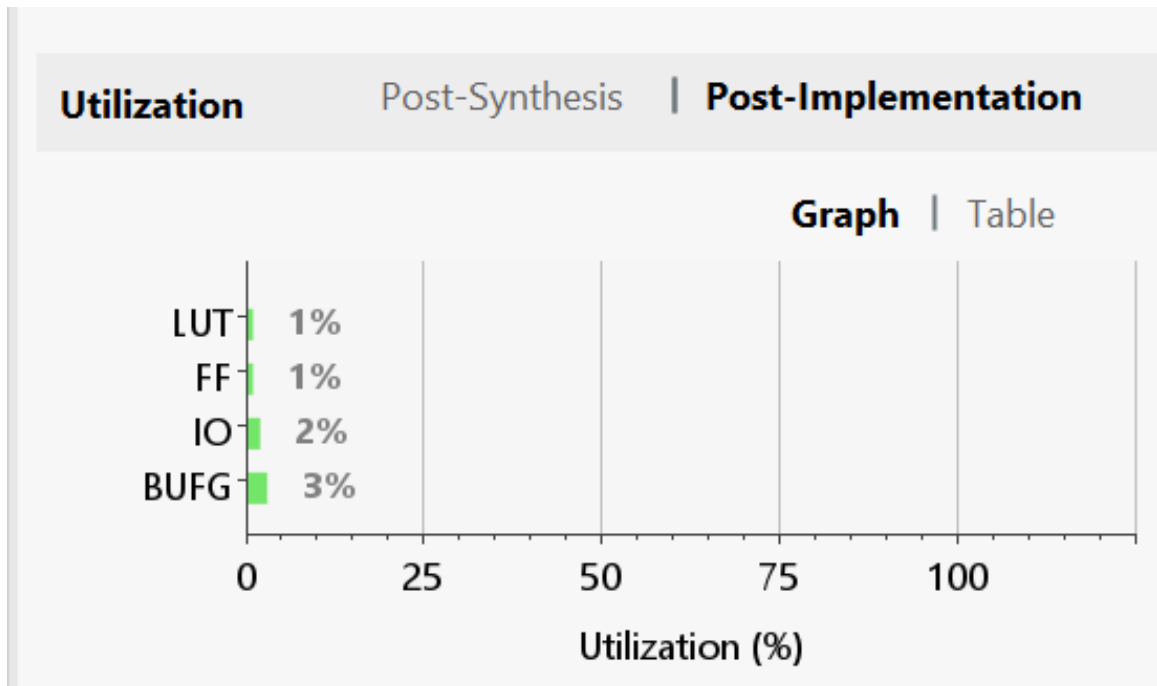


Figure 14: Device Utilization of sr flipflop

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 14.

Power Estimation

Figure 15 shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a sr flipflop circuit.

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: 0.22 W
Design Power Budget: Not Specified
Process: typical
Power Budget Margin: N/A
Junction Temperature: 26.0°C
Thermal Margin: 59.0°C (12.3 W)
Ambient Temperature: 25.0 °C
Effective θ_{JA} : 4.8°C/W
Power supplied to off-chip devices: 0 W
Confidence level: Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power

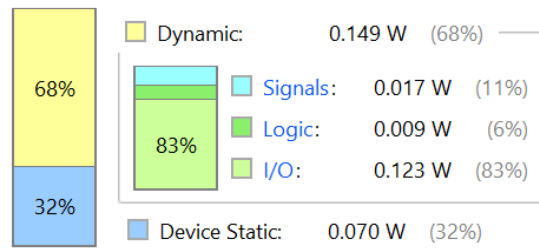


Figure 15: Power Estimation of sr flipflop

2.4 JK Flip-Flop

VHDL code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity JKFLIPFLOP is
    Port ( J : in STD_LOGIC;
          K : in STD_LOGIC;
          CLK : in STD_LOGIC;
          RESET : in STD_LOGIC;
          Q : out STD_LOGIC);
end JKFLIPFLOP;

architecture Behavioral of JKFLIPFLOP is
    signal Q_internal : STD_LOGIC := '0';
begin
    process (CLK, RESET)
    begin
        if (RESET = '1') then
            Q_internal <= '0';
        elsif (rising_edge(CLK)) then
            if (J = '1' and K = '0') then
```

```

        Q_internal <= '1';
    elsif (J = '0' and K = '1') then
        Q_internal <= '0';
    elsif (J = '1' and K = '1') then
        Q_internal <= not Q_internal;
    end if;
end if;
end process;
Q <= Q_internal;
end Behavioral;

```

Test Bench

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity JKFLIPFLOP_tb is
end JKFLIPFLOP_tb;
architecture Behavioral of JKFLIPFLOP_tb is
    component JKFLIPFLOP
        Port (
            J : in STD_LOGIC;
            K : in STD_LOGIC;
            CLK : in STD_LOGIC;
            RESET : in STD_LOGIC;
            Q : out STD_LOGIC
        );
    end component;
    signal J, K : STD_LOGIC := '0';
    signal RESET : STD_LOGIC := '0';
    signal CLK : STD_LOGIC := '0';
    signal Q : STD_LOGIC;

begin
    UUT: JKFLIPFLOP port map (
        J => J,
        K => K,
        CLK => CLK,
        RESET => RESET,
        Q => Q
    );
    CLK_process: process
    begin
        CLK <= '0';
        wait for 10 ns;
    end process;

```

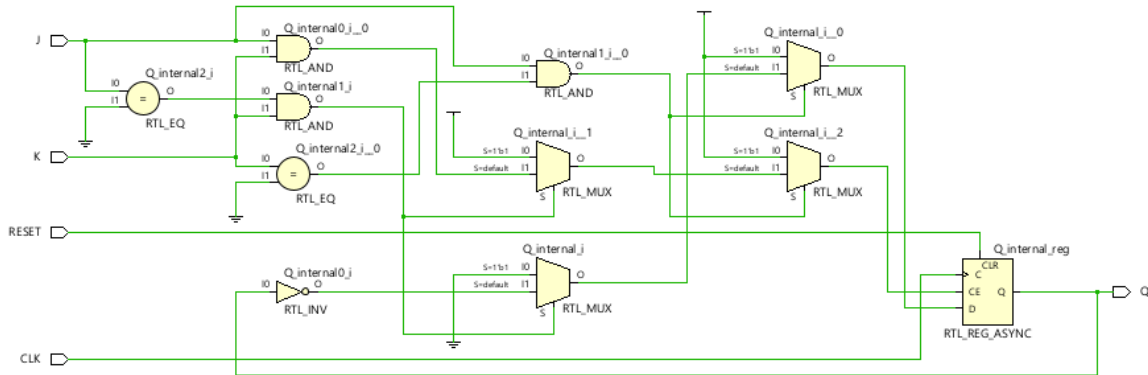


Figure 16: RTL Schematic Diagram of jk flipflop

```

        CLK <= '1';
        wait for 10 ns;
    end process;
    Stimulus: process
    begin
        RESET <= '1';
        wait for 20 ns;
        RESET <= '0';
        J <= '1'; K <= '0'; wait for 20 ns;
        J <= '0'; K <= '1'; wait for 20 ns;
        J <= '1'; K <= '1'; wait for 20 ns;
        J <= '0'; K <= '0'; wait for 20 ns;
        J <= '1'; K <= '1'; wait for 20 ns;
        wait;
    end process;
end Behavioral;

```

RTL Schematic Diagram

RTL design is used to describe the behavior of the jk flipflop as shown in figure 16

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the jk flipflop circuit as shown in figure 17

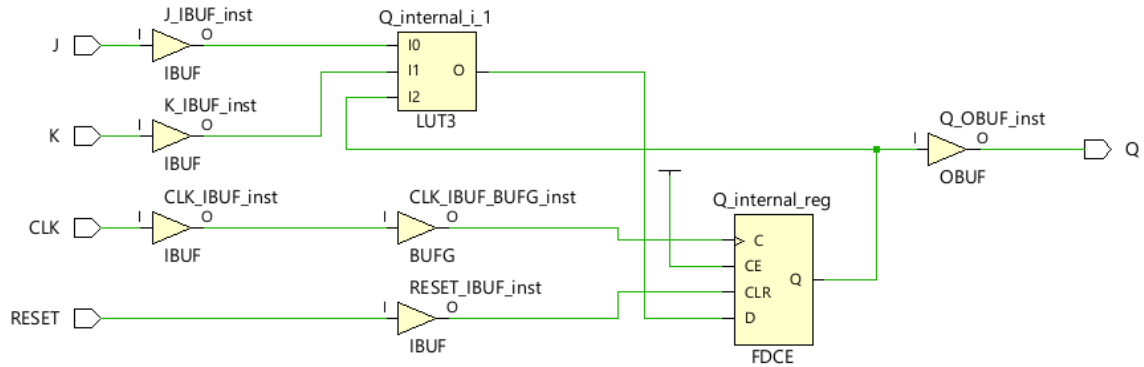


Figure 17: Synthesis Schematic Diagram of jk flipflop

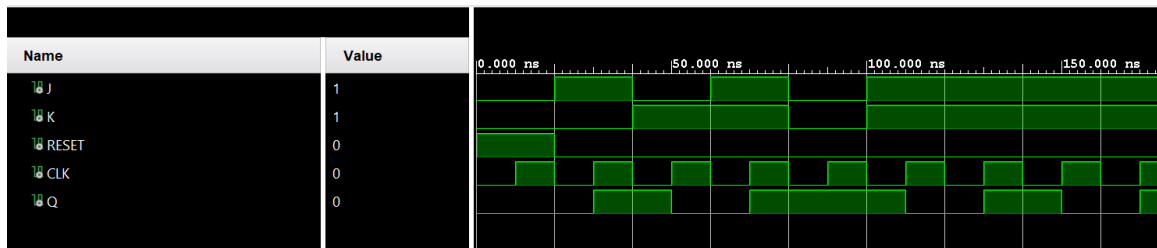


Figure 18: Timing Diagram of jk flipflop

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table of jk flipflop as shown in figure 18.

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 19.

Power Estimation

Figure 15 shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a jk flipflop circuit.

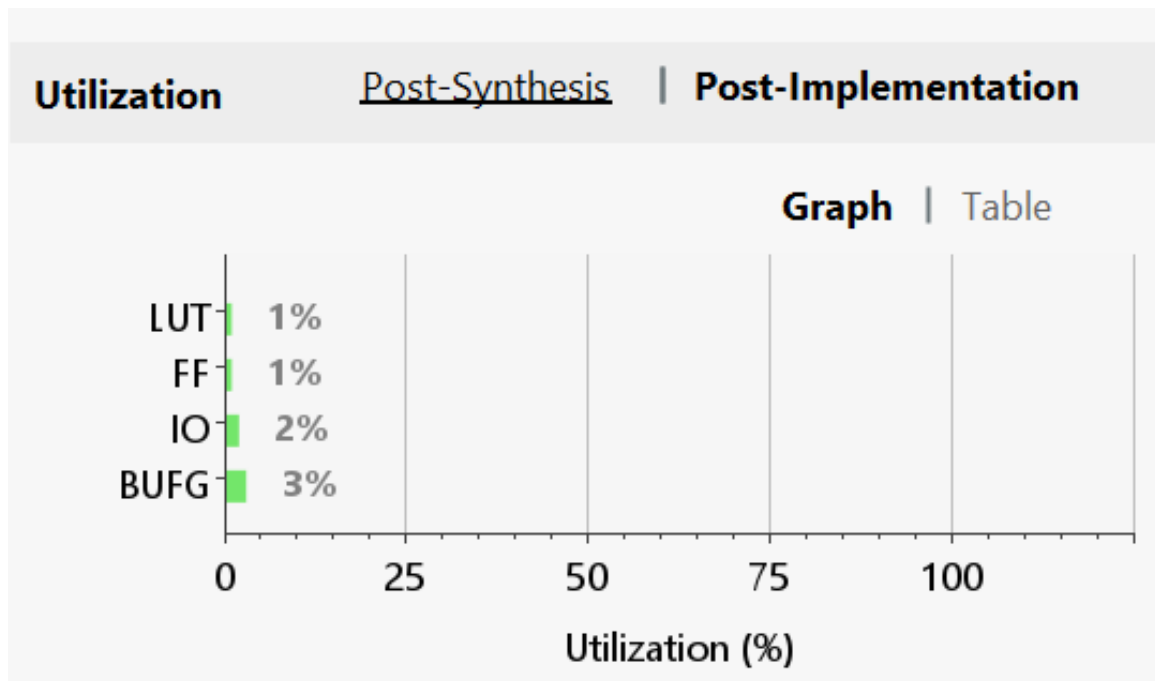


Figure 19: Device Utilization of jk flipflop

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: **0.71 W**

Design Power Budget: **Not Specified**

Process: **typical**

Power Budget Margin: **N/A**

Junction Temperature: **28.4°C**

Thermal Margin: 56.6°C (11.8 W)

Ambient Temperature: 25.0 °C

Effective θ_{JA} : 4.8°C/W

Power supplied to off-chip devices: 0 W

Confidence level: **Low**

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

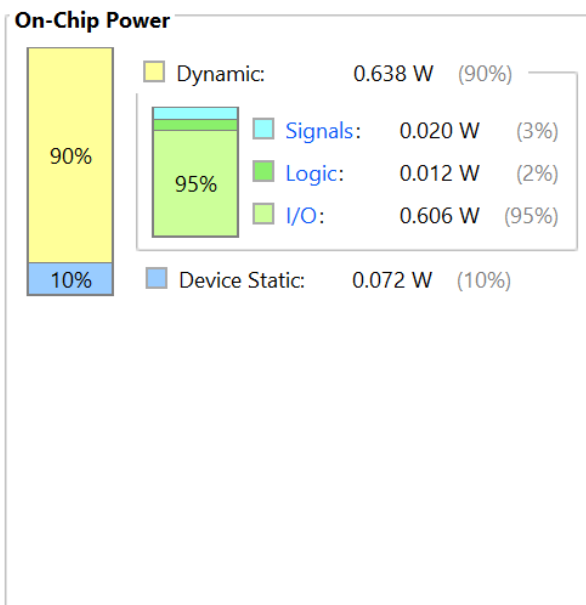


Figure 20: Power Estimation of jk flipflop

3 Conclusion

In this lab, we implemented the VHDL code for T flip-flop, D flip-flop, SR flip-flop and JK flip-flop on the Atrix-7 FPGA (part: xc7a50tcs324-1, package: csg324) using Vivado software. Each design was synthesized, simulated, and implemented, which confirmed their correct functionality. The process began with simulation to verify the timing diagrams, followed by linting to check for syntax errors. Afterward, the designs were synthesized into a hardware logic circuit, and during the implementation phase, power and timing characteristics were analyzed, displayed and studied.

Each flip-flop was implemented and verified through simulation, confirming its expected behavior and showcasing its role in sequential logic systems. This lab provided practical experience in using Vivado for digital design and strengthened the understanding of flip-flops as essential components in modern digital applications.